Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

Question 5

Part A

Tetrahedral

Part B

Yes that is correct. Since both compounds have the same types of IMFs, the higher boiling point goes to the compound w/ the larger LDF which is directly related to the larger molar mass.

Part C

Compound X will have a higher vapor pressure when heated b/c it has weaker IMF forces leading to a lower boiling point & therefore needing less energy to vaporize thus causing the higher vapor pressure.

Question 5 is continued on the next page.

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

Question 5

Part D

$$\frac{PV = nRT}{RT}$$

$$n = \frac{(2.30\,atm)(12.5\,L)}{(.08201\,L\,atm\,K^{-1}\,mol^{-1})(471\,K)}$$

$$= .744 \text{ mols}$$

Continue to Question 6.